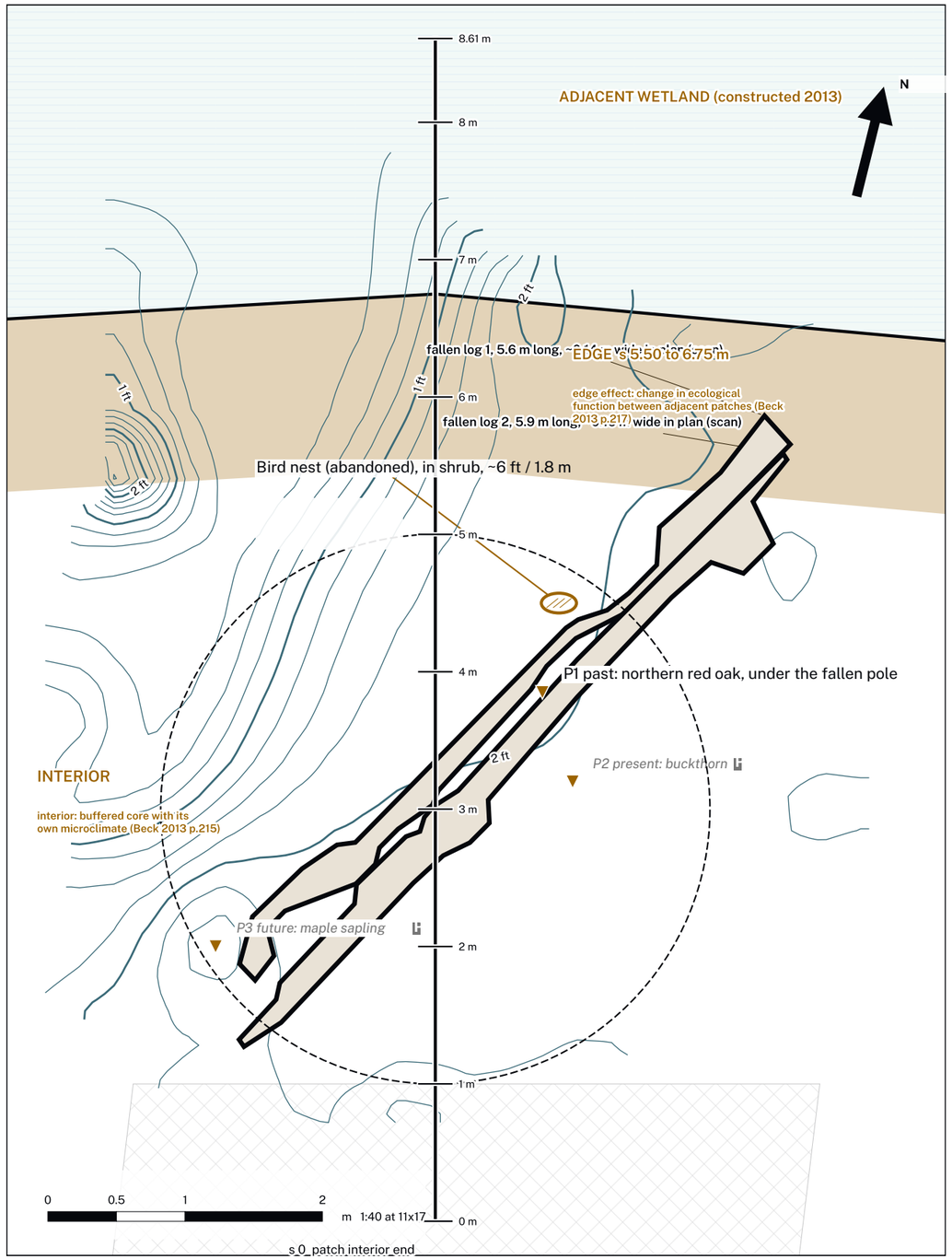
D2 Alt · Contours + Fallen Pole

Part B. Group patch on the scan footprint: contours only, on white. Same frame, scale and overlays as options A to C. Plan 1:40 at 11x17. Patch outlines are proxy until traced from the group map.

REF-01



KEY plot layers

- fallen pole and oaks, outlined from the scan (points 0.25 to 0.8 m above ground)
- ground contours 0.25 ft (scan ground grid)
- index contour 1 ft, labelled (ft, scan z datum; relief 0.1 to 2.8 ft)
- edge: functional band 1.25 m (s 5.50 to 6.75)
- group patch boundary (proxy outline)
- adjacent patches, spec A1 classes (proxy outlines)
- transect, ticks every 1 m, s 0 at the patch interior
- fixed-radius plot r 2.0 m (proxy)
- P1 past (oak under the fallen pole) · P2 present · P3 future
- habitat feature: bird nest (abandoned), position

12.3 m measured (scan / field)

term (reading, page)

estimated

P = estimated, not measured.

SITE NOTE

Adjacent wetland: constructed stormwater wetland, 2013, DCR / Cambridge / MWRA

3.4 ac, completed Oct 2013 (MWRA CSO program). Sources: MWRA news release 10/15/2013, mwra.com/01news/2013/101513-eea-wetland-cambridge.html; Hatch project page, hatch.com/Projects/Infrastructure/Alewife-Stormwater-Wetland

TERMS (readings)

patch: an area that differs from what surrounds it (Beck 2013 p.211)
edge effect: change in ecological function between adjacent patches (Beck 2013 p.217)
disturbance: event that disrupts structure (Beck 2013 p.179)

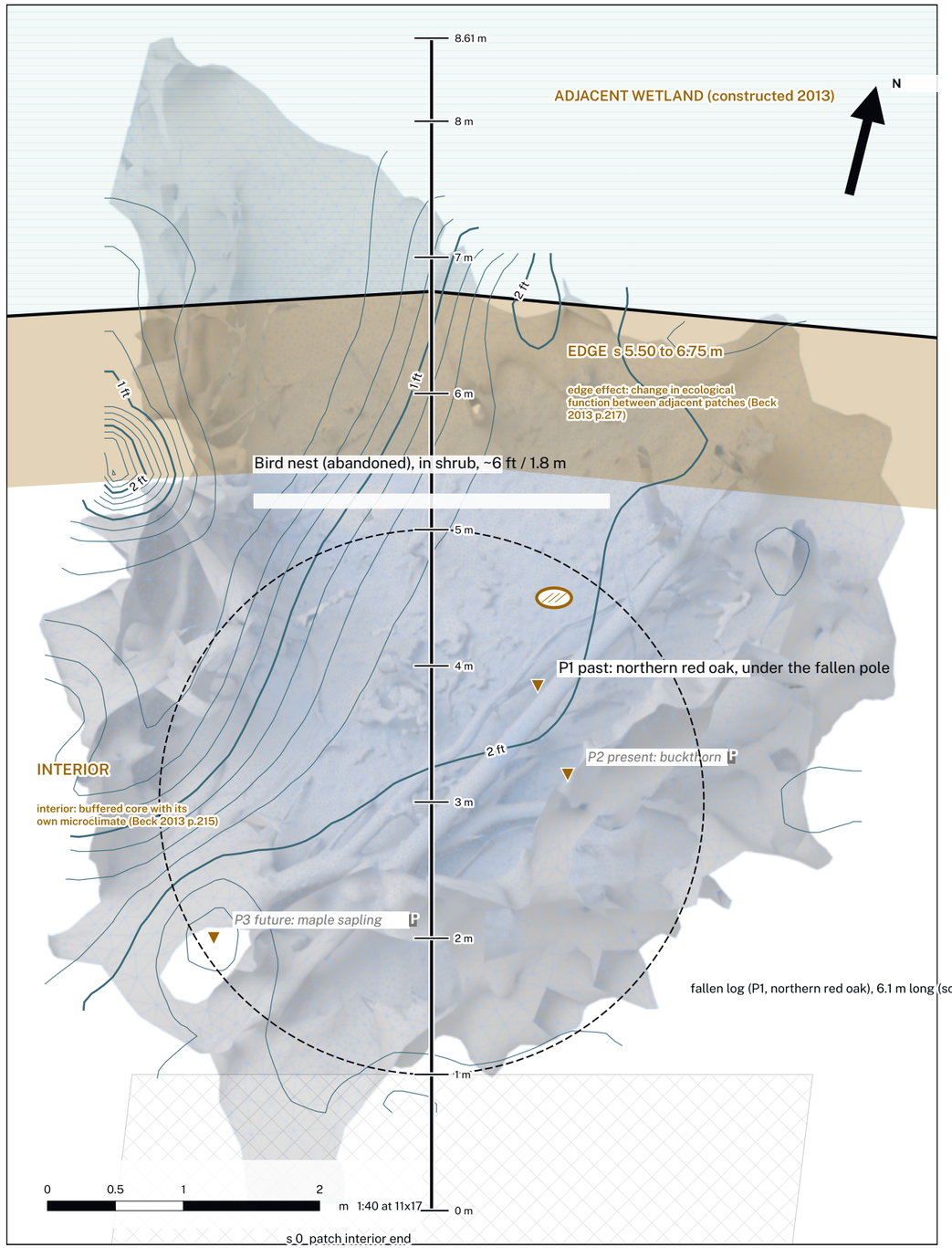
PLANT TABLE the group's plant catalog · companions

common	Latin	form / str.	patch	wetl. ind.	origin / MA	role
Japanese knotweed	*Reynoutria japonica*	herb, tall / U	bank thicket, disturbed edge	FACU	E Asia; invasive, prohibited in MA	edge, disturba.
European buckthorn	*Rhamnus cathartica*	shrub / M	canopy gap, bank thicket	FAC	Europe; invasive, prohibited in MA	P2 present
Goldenrod	*Solidago spp.*	herb / G	edge, openings	FACU to FACW	native	
Sugar maple	*Acer saccharum*	tree / O (sapling U)	floodplain woods, gap saplin.	FACU	native	P3 future
Domestic apple	*Malus domestica*	tree / M	floodplain woods edge	UPL, confirm	Central Asia; non-native, not listed	
Northern red oak	*Quercus rubra*	tree / O	under the fallen pole	FACU	native	P1 past
Oak, caught at Y-bran.	*Quercus sp.*	tree / O	under the pole		native	companion
Fallen utility pole	*not a plant*	legacy wood / G	across the plot			companion
Bird's nests (2, abando.	*habitat feature*	shrub, ~6 ft	s 4 to 5 m; 2nd not recorded			habitat

D2 Alt · Contours + Mesh

REF-02

Part B. Group patch on the scan footprint: Polycam mesh wireframe, top view, ortho, lightened 45 %. Same frame, scale and overlays as options A to C. Plan 1:40 at 11x17. Patch outlines are proxy until traced from the group map.



KEY plot layers

- base: Polycam mesh wireframe, top view, ortho, lightened 45 %
- ground contours 0.25 ft (scan ground grid)
- index contour 1 ft, labelled (ft, scan z datum; relief 0.1 to 2.8 ft)
- edge: functional band 1.25 m (s 5.50 to 6.75)
- group patch boundary (proxy outline)
- adjacent patches, spec A1 classes (proxy outlines)
- transect, ticks every 1 m, s 0 at the patch interior
- fixed-radius plot r 2.0 m (proxy)
- P1 past (oak under the fallen pole) · P2 present · P3 future
- habitat feature: bird nest (abandoned), position

12.3 m measured (scan / field)

term (reading, page)

estimated

P = estimated, not measured.

SITE NOTE

Adjacent wetland: constructed stormwater wetland, 2013, DCR / Cambridge / MWRA

3.4 ac, completed Oct 2013 (MWRA CSO program). Sources: MWRA news release 10/15/2013, mwra.com/01news/2013/101513-eea-wetland-cambridge.html; Hatch project page, hatch.com/Projects/Infrastructure/Alewife-Stormwater-Wetland

TERMS (readings)

patch: an area that differs from what surrounds it (Beck 2013 p.211)
edge effect: change in ecological function between adjacent patches (Beck 2013 p.217)
disturbance: event that disrupts structure (Beck 2013 p.179)

PLANT TABLE the group's plant catalog · companions

common	Latin	form / str.	patch	wetl. ind.	origin / MA	role
Japanese knotweed	*Reynoutria japonica*	herb, tall / U	bank thicket, disturbed edge	FACU	E Asia; invasive, prohibited in MA	edge, disturba.
European buckthorn	*Rhamnus cathartica*	shrub / M	canopy gap, bank thicket	FAC	Europe; invasive, prohibited in MA	P2 present
Goldenrod	*Solidago spp.*	herb / G	edge, openings	FACU to FACW	native	
Sugar maple	*Acer saccharum*	tree / O (sapling U)	floodplain woods, gap saplin.	FACU	native	P3 future
Domestic apple	*Malus domestica*	tree / M	floodplain woods edge	UPL, confirm	Central Asia; non-native, not listed	
Northern red oak	*Quercus rubra*	tree / O	under the fallen pole	FACU	native	P1 past
Oak, caught at Y-bran.	*Quercus sp.*	tree / O	under the pole		native	companion
Fallen utility pole	*not a plant*	legacy wood / G	across the plot			companion
Bird's nests (2, abando.	*habitat feature*	shrub, ~6 ft	s 4 to 5 m; 2nd not recorded			habitat

Species Table (full)

REF-03

Primary species from the group's plant catalog, with its notes (lightly edited). Companions: the oak (P1), the pole, the nests. Earlier tentative photo IDs below.

PRIMARY + COMPANIONS

common	Latin	form / str.	patch	wetl. ind.	origin / MA	role
Japanese knotweed	*Reynoutria japonica*	herb, tall / U	bank thicket, disturbed edge	FACU	E Asia; invasive, prohibited in MA	edge, disturban.
European buckthorn	*Rhamnus cathartica*	shrub / M	canopy gap, bank thicket	FAC	Europe; invasive, prohibited in MA	P2 present
Goldenrod	*Solidago spp.*	herb / G	edge, openings	FACU to FACW	native	
Sugar maple	*Acer saccharum*	tree / O (sapling U)	floodplain woods, gap saplings	FACU	native	P3 future
Domestic apple	*Malus domestica*	tree / M	floodplain woods edge	UPL, confirm	Central Asia; non-native, not listed	
Northern red oak	*Quercus rubra*	tree / O	under the fallen pole	FACU	native	P1 past
Oak, caught at Y-branch	*Quercus sp.*	tree / O	under the pole		native	companion
Fallen utility pole	*not a plant*	legacy wood / G	across the plot			companion
Bird's nests (2, abandoned)	*habitat feature*	shrub, ~6 ft	s 4 to 5 m; 2nd not recorded			habitat

NOTES (the group's plant catalog)

Japanese knotweed (Reynoutria japonica): Dense stands to about 10 ft that shade out natives; spreads by long rhizomes; rivers, roadsides, rail corridors, disturbed urban sites.

European buckthorn (Rhamnus cathartica): Introduced as a hedge plant; birds eat and spread the black berries; dense shade suppresses native understory.

Goldenrod (Solidago spp.): Many species in MA; blooms late summer to fall; key nectar source for bees and butterflies before winter.

Sugar maple (Acer saccharum): Signature New England tree; source of maple syrup; can live 300+ years; food and habitat for birds and squirrels.

Domestic apple (Malus domestica): Spring blossoms for pollinators; fruit for birds, mammals, people; feral apples common in old New England fields.

Indicator: USACE National Wetland Plant List, Northcentral and Northeast. MA invasive: MDAR Prohibited Plant List. Silver maple (A. saccharinum, FACW) is the floodplain maple expected here; confirm against the sugar maple ID. Hypothesis: knotweed and buckthorn thrive in disturbed openings such as the pole-fall gap.

~~EARLIER TENTATIVE PHOTO IDs (scan keyframes, not confirmed; below the catalog, dropped where they conflict)~~

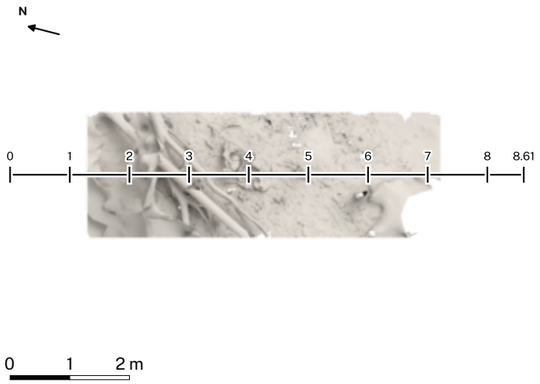
common	Latin	form	stratum	where	status
Glossy buckthorn	*Frangula alnus*	shrub	M	interior (s 1 to 5), plot	tentative (photo)
Dogwood	*Cornus sp.*	shrub	M	interior (approx, from photo)	tentative (photo)
Oriental bittersweet	*Celastrus orbiculatus*	vine	M	interior (approx, from photo)	tentative (photo)
Red maple	*Acer rubrum*	tree (seedlings)	U	interior (plot), edge (s 5)	estimated, not observed
Ash seedling	*Fraxinus sp.*	tree (seedling)	U	interior (approx, from photo)	tentative (photo)
Virginia creeper	*Parthenocissus quinquefolia*	vine	U	interior (approx, from photo)	tentative (photo)
Poison ivy	*Toxicodendron radicans*	vine	U	interior (plot)	estimated, not observed
Bluegrass	*Poa sp.*	graminoid	G	interior (s 0)	estimated, not observed
Moss on the pole	*Bryophyta*	moss	G	interior (on the pole)	tentative (photo)
Willow	*Salix sp.*	shrub or tree	M	edge (s 6)	estimated, not observed
Black cherry	*Prunus serotina*	tree (bark only)	M	wetland side (approx, from photo)	tentative (photo)
Jewelweed	*Impatiens capensis*	herb	G	wetland (s 7)	estimated, not observed
Sedge	*Carex sp.*	graminoid	G	wetland (s 8), plot	estimated, not observed

Scan Atlas

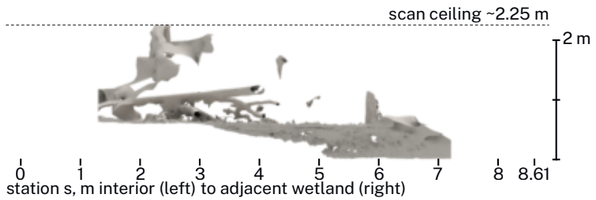
REF-04

Polycam mesh of the transect (10_23_2026.zip, OBJ + baked texture), trimmed to a 5.91 x 2.10 m rectangle on the transect, in three shadings and three views. Top and side are orthographic in the transect frame, side looks across the transect. s = 0 at patch interior.

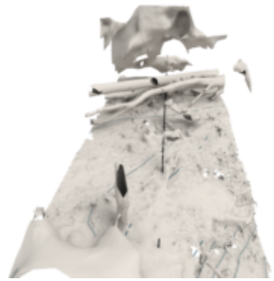
TOP ortho 1:100
CLAY flat warm grey, studio light + AO



SIDE ortho 1:100, looking across the transect



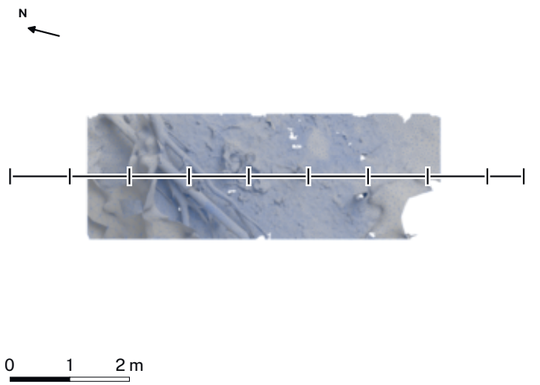
PERSPECTIVE CAM_persp_A

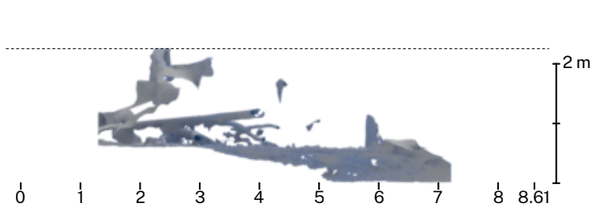


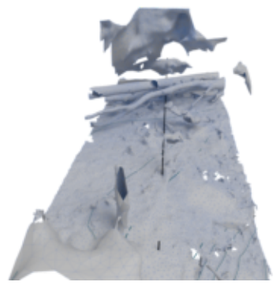
GROUND CONTOURS over the texture top view, ortho 1:75



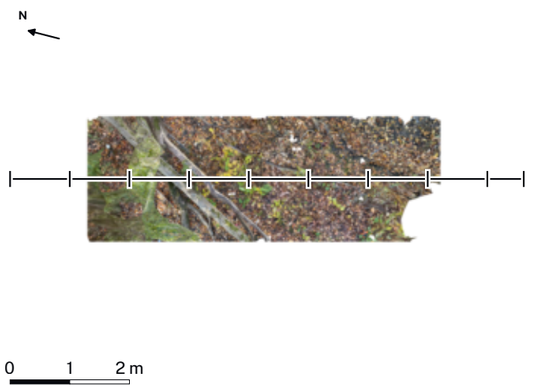
WIRE shaded grey, mesh edges in water blue







TEXTURE baked Polycam texture, unlit (no added light)







perspective, no scale (22 mm lens, 36 mm sensor). from the adjacent wetland end, looking back at the fallen pole

KEY

transect on the PCA axis of the scan, ticks every 1 m (stations)

ground contours 0.05 m, index 0.25 m labelled (m, scan z)

scan ceiling: the mesh stops at 2.25 m

scan ceiling ~2.25 m, canopy not captured.

Contours are cut from the ground surface grid (5th percentile z per 0.25 m cell), not the mesh, so stems and logs do not ring.

North is not known: Polycam's frame is local. Arrow assumes scan +y = north .

Mesh trimmed to a 5.91 x 2.10 m rectangle on the transect: 51 084 faces (full scan 100 000), one 4K colour texture. Rendered in Blender 4.4.1, CYCLES CPU from 01_src/blender/ett3-A3_scan_v01.blend.

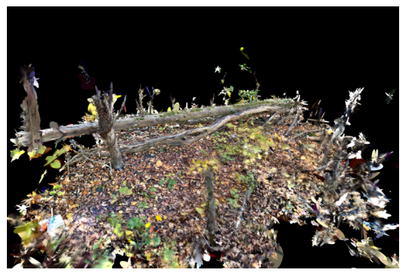
Full-size views (3000 px, transparent PNG): 00_share/views/.

Measured from scan only; no estimateds on this sheet.

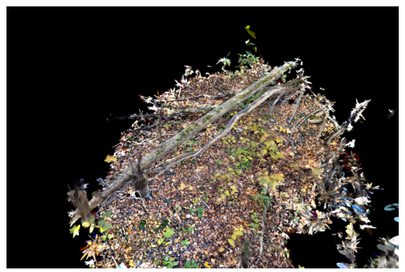
Splat Atlas

Gaussian splat trained from 544 keyframes (COLMAP + Brush), aligned to the LiDAR scan (2.1 cm RMSE). 23 shots. Wetland shots weak: few views of open water.

REF-05



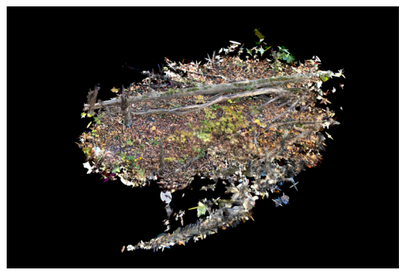
CAM persp A



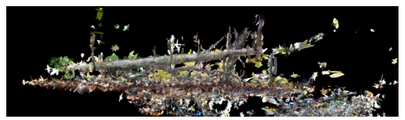
CAM persp B



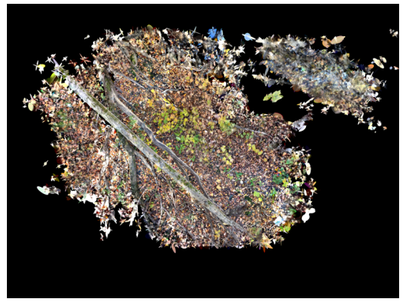
CAM persp C



CAM persp D



CAM side



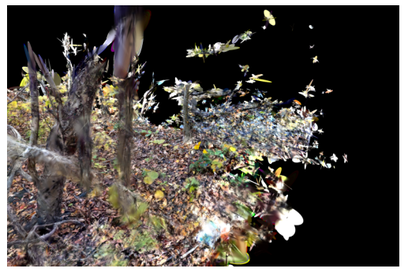
CAM top



ZOOM bank close



ZOOM bank mid



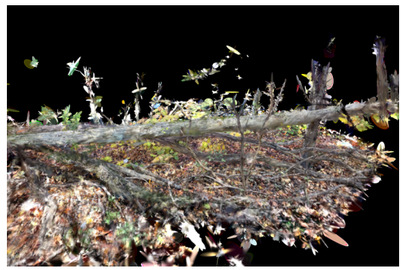
ZOOM bank wide



ZOOM pole close



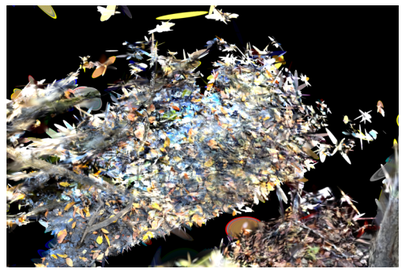
ZOOM pole mid



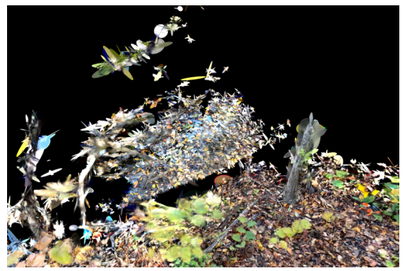
ZOOM log wide



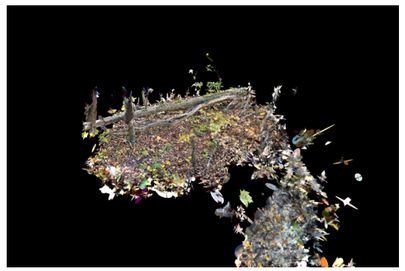
ZOOM wetland close



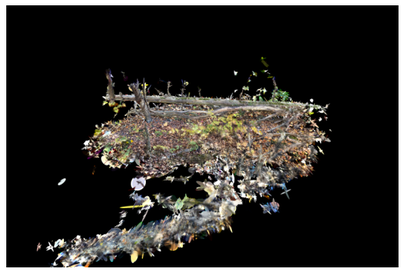
ZOOM wetland mid



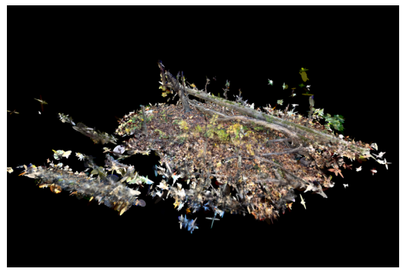
ZOOM wetland wide



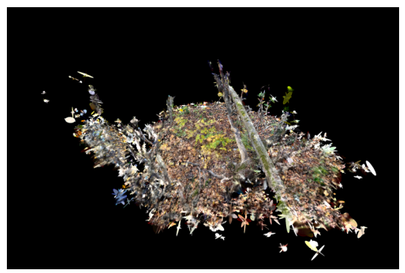
ORBIT 000



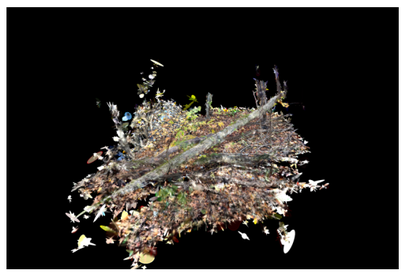
ORBIT 045



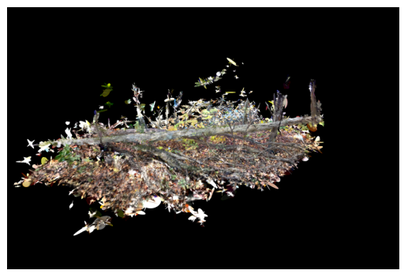
ORBIT 090



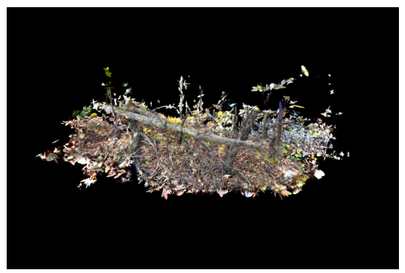
ORBIT 135



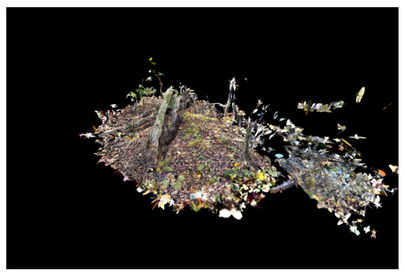
ORBIT 180



ORBIT 225



ORBIT 270



ORBIT 315

Data

The data file behind D2 and D3, one row per station. Grey italic + P = proxy. Replace with field data: see README_group_v01.md.

REF-06

station s, m	veg ht max, m	strata	openness, %	cover, %	woody, %	species, n	moisture, % VWC	soil pH	compaction, kPa	litter, cm	dominant species	source
0	0.31	ground	*30*	*35*	*5*	*3*	*18*	*6.4*	*1400*	*1.0*	*Poa sp*	proxy 9 · scan 2
1	1.11	ground|understory	*22*	*70*	*55*	*5*	*20*	*6.3*	*900*	*3.5*	*Frangula alnu*	proxy 9 · scan 2
2	1.61	ground|understory	*18*	*85*	*70*	*6*	*22*	*6.2*	*750*	*5.0*	*Frangula alnu*	proxy 9 · scan 2
3	1.47	ground|understory	*18*	*80*	*60*	*6*	*24*	*6.2*	*700*	*5.5*	*Frangula alnu*	proxy 9 · scan 2
4	1.69	ground|understory	*20*	*80*	*65*	*5*	*27*	*6.1*	*700*	*5.0*	*Frangula alnu*	proxy 9 · scan 2
5	1.95	ground|understory	*25*	*75*	*55*	*6*	*31*	*6.0*	*650*	*4.0*	*Acer rubrun*	proxy 9 · scan 2
6	1.07	ground|understory	*40*	*60*	*30*	*7*	*42*	*6.3*	*500*	*2.0*	*Salix sp*	proxy 9 · scan 2
7	0.29	ground	*70*	*85*	*5*	*5*	*58*	*6.6*	*300*	*0.5*	*Impatiens capensi*	proxy 9 · scan 2
8	0.18	ground	*80*	*90*	*0*	*4*	*62*	*6.7*	*250*	*0.0*	*Carex sp*	proxy 9 · scan 2

ZONES (scan)

s 0.00 to 1.00 m	interior gap (bare ground / exposed soil)
s 1.00 to 5.50 m	interior (trees / woody)
s 5.50 to 6.75 m	edge (trees / woody)
s 6.75 to 8.61 m	adjacent patch (wetland)
transect length	8.61 m

PLOT

centre_s_m	*3.0 m*
radius_m	*2.0 m*
cover: Frangula alnus	*45 %*
cover: Acer rubrum (seedlings)	*10 %*
cover: Carex sp.	*15 %*
cover: Toxicodendron radicans	*5 %*

PLANTS

P1 past	*Salix sp*
position x y, m	*3.5 6.3*
P2 present	*Frangula alnu*
position x y, m	*1.0 3.2*
P3 future	*Acer rubrum (seedlin*
position x y, m	*-1.6 2.0*

KEY

12.3 m measured (scan / field)

Concept: author year

estimated

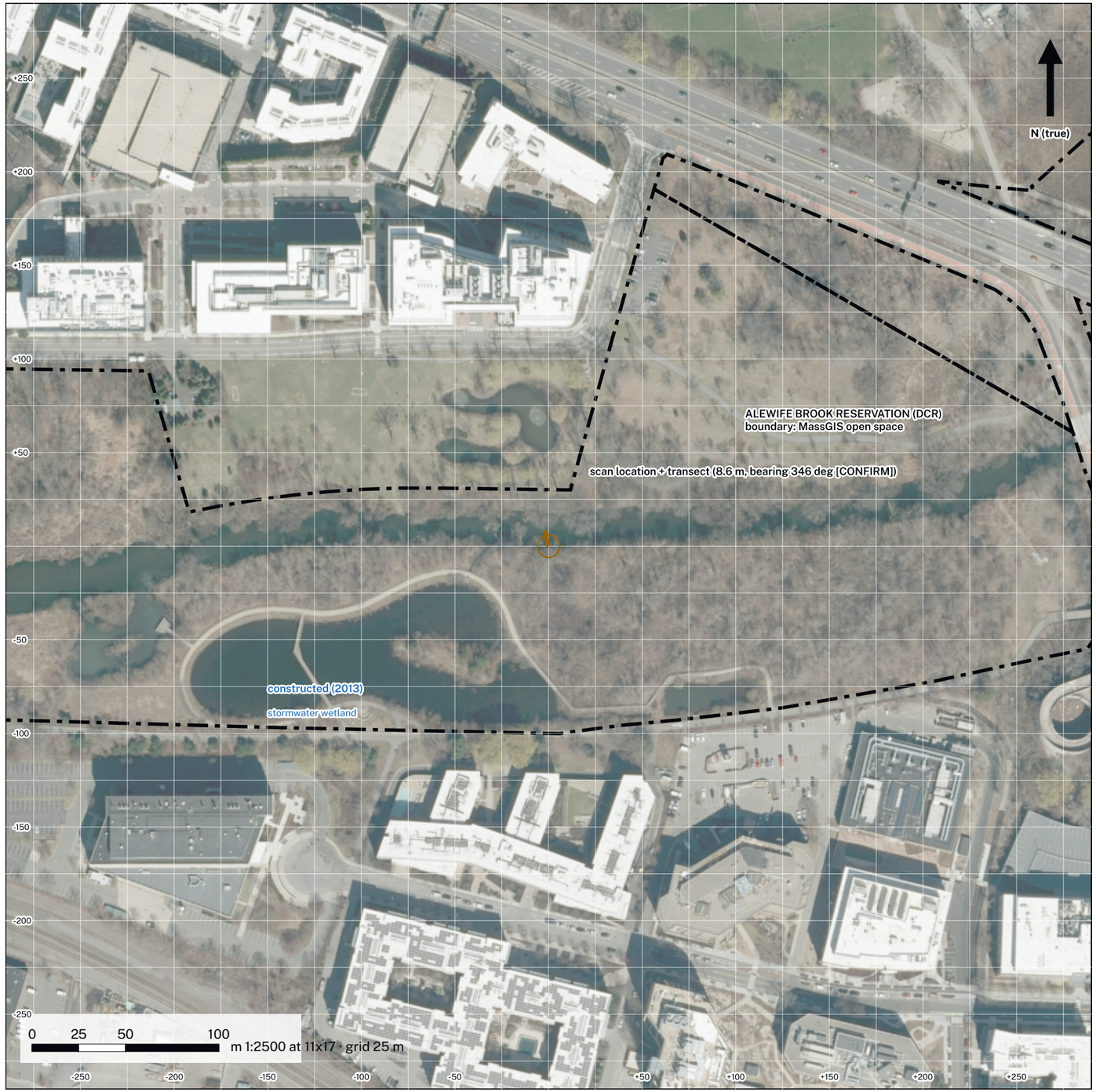
P = estimated, not measured.

see README_group_v01.md]

Blank Base Plan

REF-07

Blank base for the group's patch map: MassGIS 2023 leaf-off aerial at 1:2500, reservation boundary, scan location. No patch lines: trace patches by hand, in Illustrator (empty layer per class) or in My Maps (KMZ, empty folder per class).



KEY patch classes (spec A1), swatches empty: fill in your colour

☐	**F-D**	woody, deciduous: Dm mature / Dy young
☐	**F-E**	woody, evergreen
☐	**S**	shrub
☐	**H**	upland herbaceous
☐	**W**	wetland
☐	**B**	bare ground
☐	**P**	paving
☐	**O**	open water
☐	**A**	tone anomaly

____ certain edge --- unsure edge (dashed)

Rules: draw an edge where tone or texture changes over ~10 m; minimum patch ~100 m2.

SITE NOTE

Adjacent wetland: constructed stormwater wetland, 2013, DCR / Cambridge / MWRA

3.4 ac, completed Oct 2013 (MWRA CSO program). Sources: MWRA news release 10/15/2013, mwra.com/01news/2013/101513-eea-wetland-cambridge.html; Hatch project page, hatch.com/Projects/Infrastructure/Alewife-Stormwater-Wetland

FIELD NOTES

Aerial: MassGIS orthos2023, spring 2023 leaf-off, 15 cm (tiles.arcgis.com/.../orthos2023/MapServer). Boundary: MassGIS Protected and Recreational OpenSpace (gis.eea.mass.gov/server/rest/services/Protected_and_Recreational_OpenSpace_Polygons), Alewife Brook Reservation, DCR. Scan location 42.396871, -71.146994.